# CONNIFY: CONNECT SAFELY

**A PROJECT REPORT**

***Submitted by***


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# K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY

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## BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**CONNIFY: CONNECT SAFELY**" has been carried out by the students **GANGASH S, GIRIJESH R, GODFREY T R, GRISH NARAYANAN S**. The work reported here is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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# ABSTRACT

Safe communication between strangers in physical proximity is a growing need in modern urban life, yet existing location-aware applications depend on continuously sending raw GPS coordinates to centralized servers, exposing users to tracking, stalking, and location profiling. Standard hashing also fails on GPS data because sensor drift changes the hash completely. To overcome these limitations, Connify is developed as a decentralized, zero-knowledge spatial social and communication initialization platform using React Native, TypeScript, and a Fastify, Prisma, and MongoDB backend. The system introduces the SHARP Protocol, which converts a nine-cell spatial neighborhood into a 1024-bit Bloom filter and applies BCH(15,7) error correction over the Galois Field GF(2^4), so that only 8-bit parity syndromes and blinded SHA-256 hashes reach the server. Nearby peers correct GPS drift on their own device, verify proximity, and complete a hardware-backed Ed25519 QR handshake that issues a short-lived Just-In-Time trust capsule. Real-time Socket.IO communication, an offline fallback, a behavioral risk engine, a hash-chained audit ledger, and a personal safety suite make Connify, now published on the Google Play Store, a private, secure, and practical solution for connecting strangers.

**Keywords:** Zero-Knowledge Proximity, SHARP Protocol, BCH Error Correction, Galois Field, Bloom Filter, Ed25519, React Native, Fastify, Hash-Chained Audit Ledger

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**SIGNATURE**

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# LIST OF ABBREVIATIONS

| | | |
|---|---|---|
| API | - | Application Programming Interface |
| BCH | - | Bose-Chaudhuri-Hocquenghem |
| BLE | - | Bluetooth Low Energy |
| FCM | - | Firebase Cloud Messaging |
| FNV | - | Fowler-Noll-Vo |
| GF | - | Galois Field |
| GPS | - | Global Positioning System |
| JIT | - | Just-In-Time |
| JSON | - | JavaScript Object Notation |
| JWT | - | JSON Web Token |
| mDNS | - | Multicast Domain Name System |
| OS | - | Operating System |
| P2P | - | Peer-to-Peer |
| QR | - | Quick Response |
| REST | - | Representational State Transfer |
| SDK | - | Software Development Kit |
| SHA | - | Secure Hash Algorithm |
| SHARP | - | Syndrome-Based Error-Correction Spatial Matching |
| SMS | - | Short Message Service |
| UI | - | User Interface |
| VoIP | - | Voice over Internet Protocol |
| ZK | - | Zero-Knowledge |